

Dynamics of Computation in the Brain

76908

Solution EX #5

Barak Haim

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1 Numerical simulation of the associative memory model

1.1 Stability of Memory Patterns for Large- β

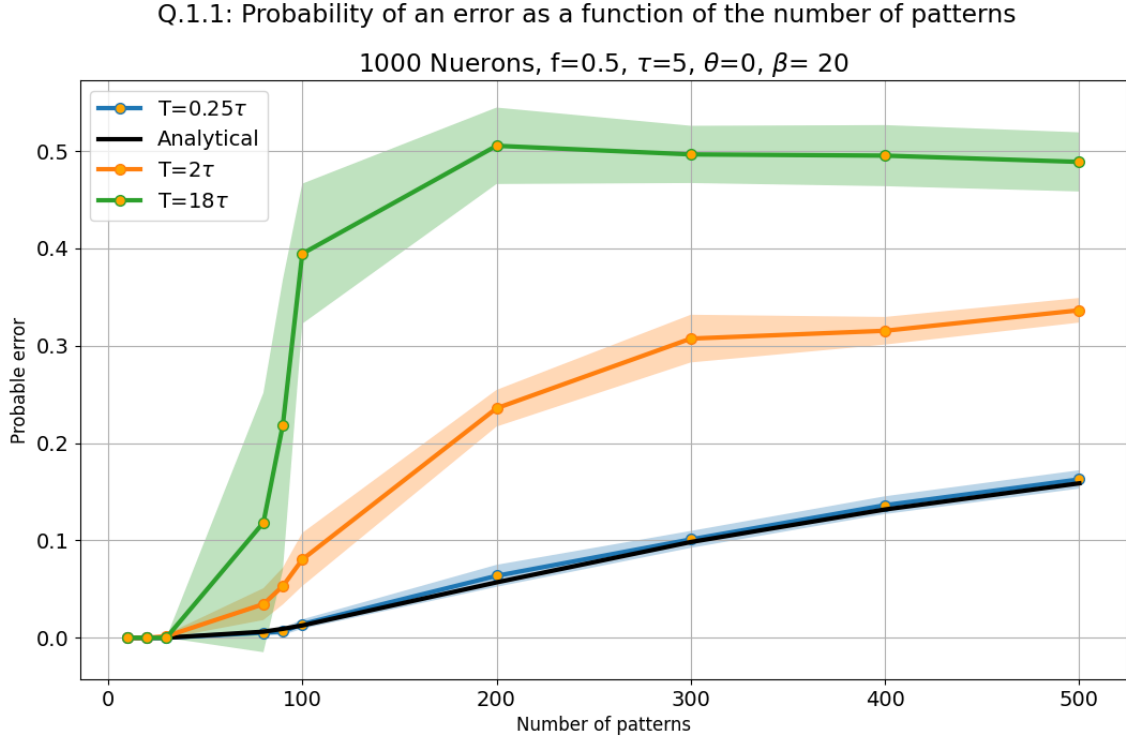


Figure 1: Stability of memory patterns as a function of the number of patterns

In the figure we see how the mean probable error changes as a number of memory patterns for simulation times $\frac{\tau}{4}$, 2τ and 18τ . Shaded areas mark the standard deviation. We used a finite (and small) β . For our purpose it may be almost the largest value we can use but for a long time-scale, it may not be sufficient to simulate $\beta \rightarrow \infty$ and prevent errors. We see that as P grows, so does the probability for an error, the error variance and the distribution of the signal to noise ratio which ultimately tells us the probability of a pattern to become a FP in the network.

1.2 Overlap Dynamics and β - Dependence

1.2.1 Overlap score in time for different β 's

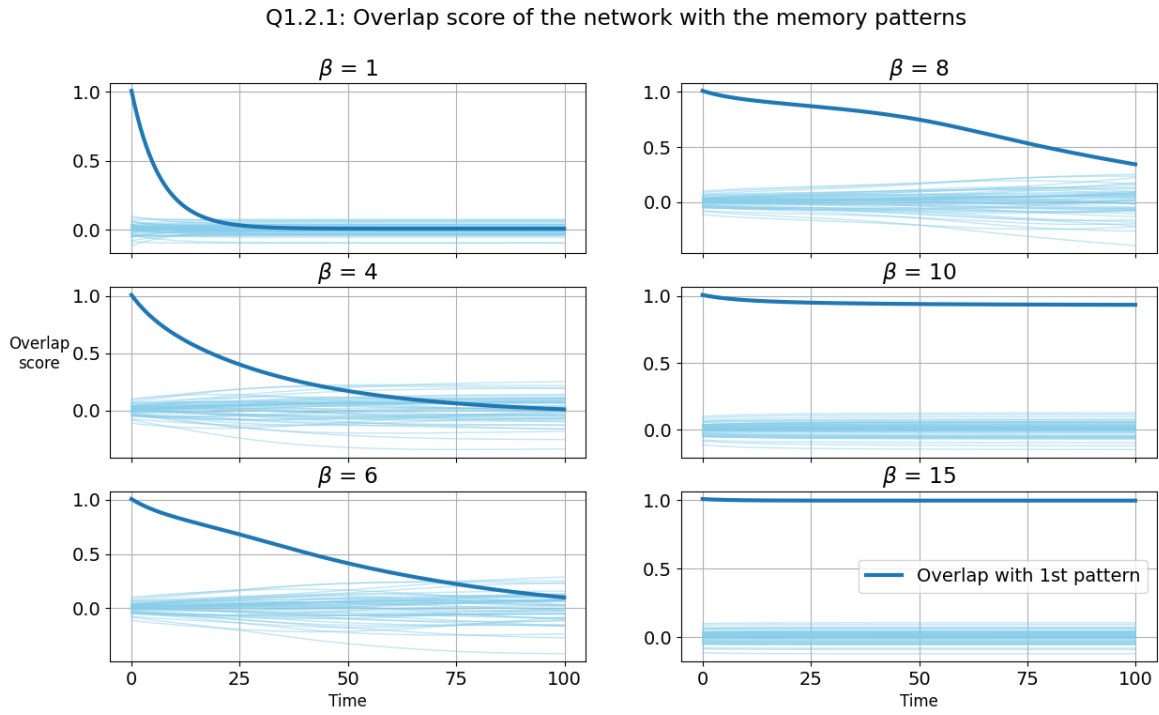


Figure 2: Overlap score in time

1.2.2 Average final value of the overlap as a function of β

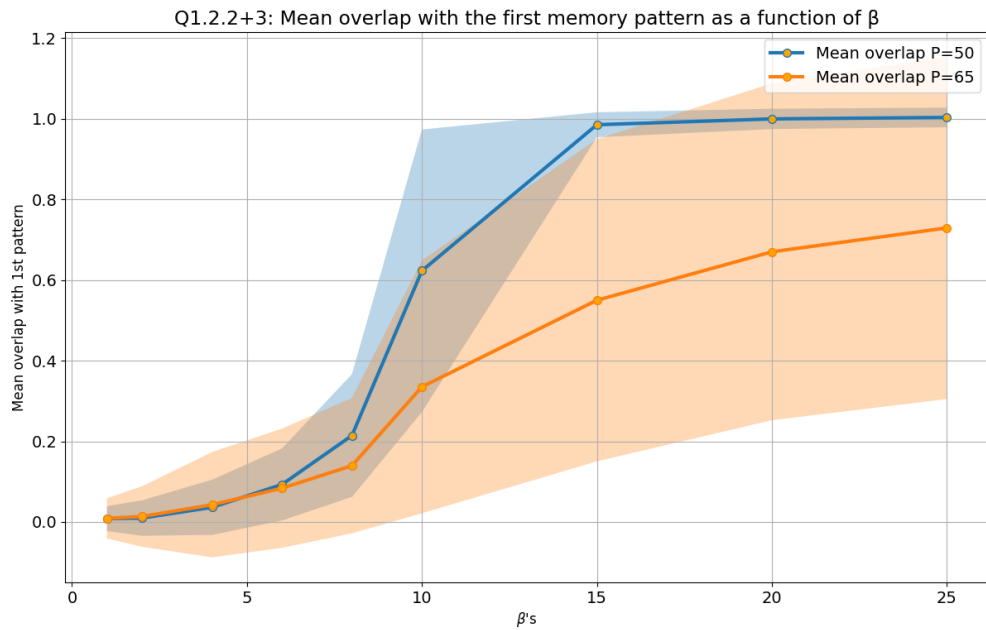


Figure 3: Mean overlap VS β

1.3 Associative Memory Recall

Q1.3: Mean error rate and correct recalls as a function of the fraction of errors

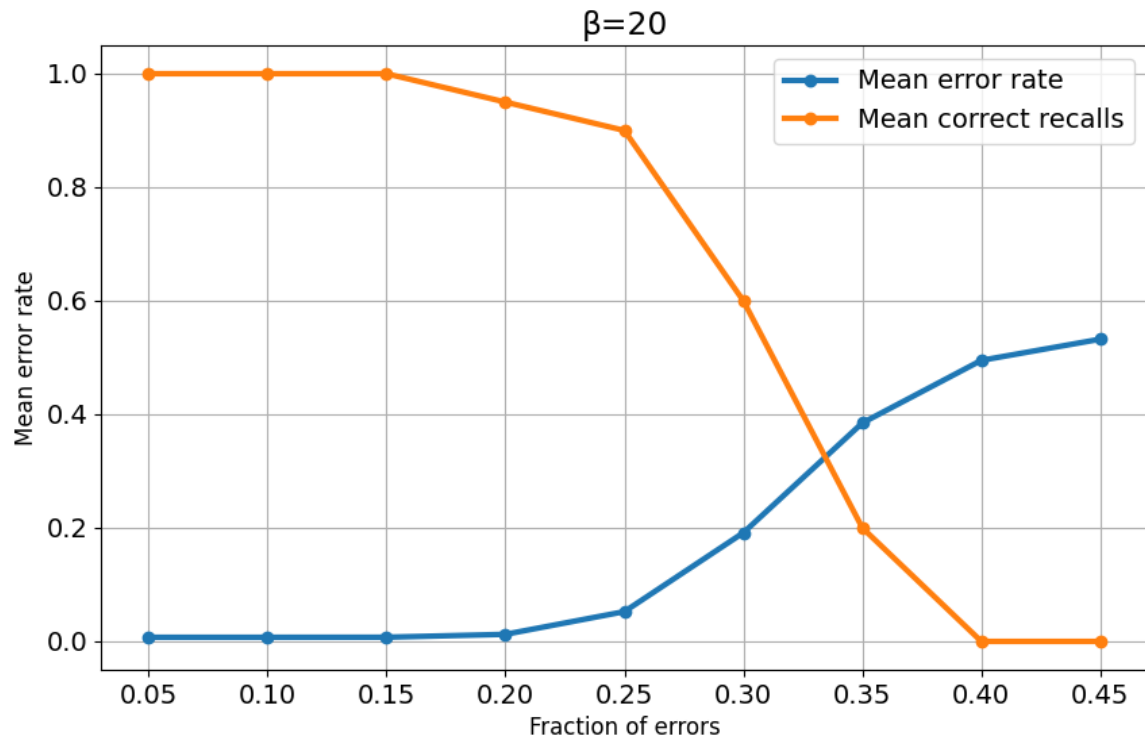


Figure 4: Recall rate and error rate $\beta = 20$

Q1.3: Mean error rate and correct recalls as a function of the fraction of errors

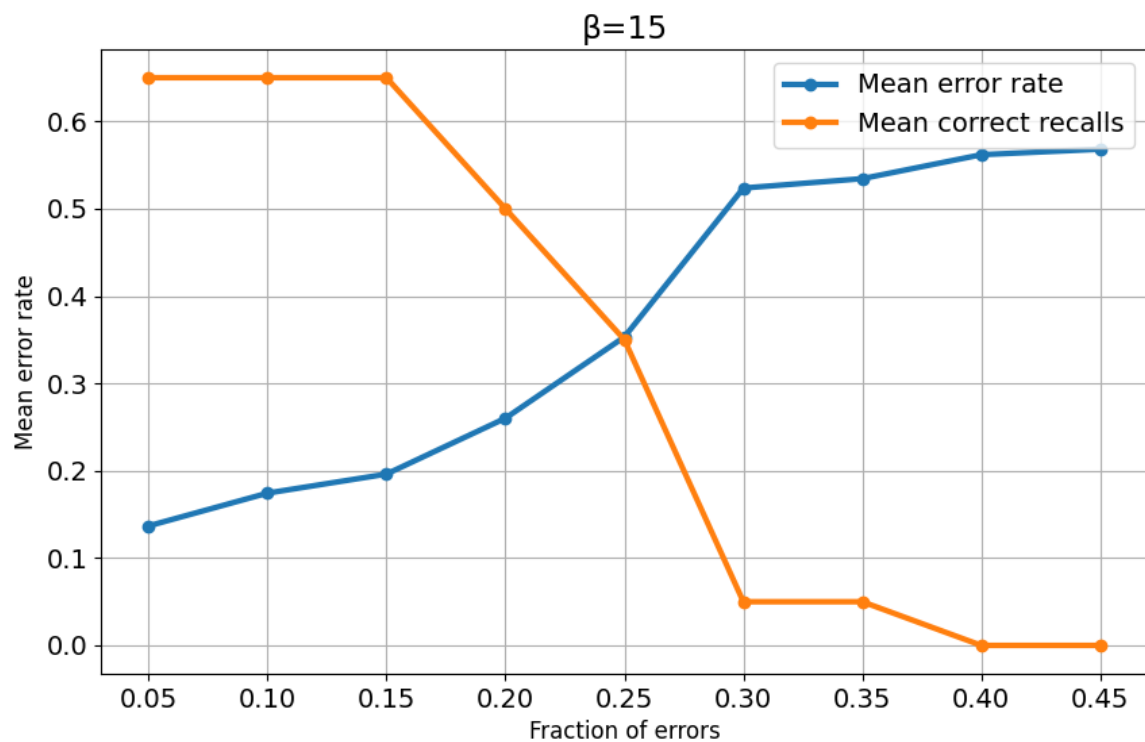


Figure 5: Recall rate and error rate $\beta = 15$

Q1.3: Mean error rate and correct recalls as a function of the fraction of errors

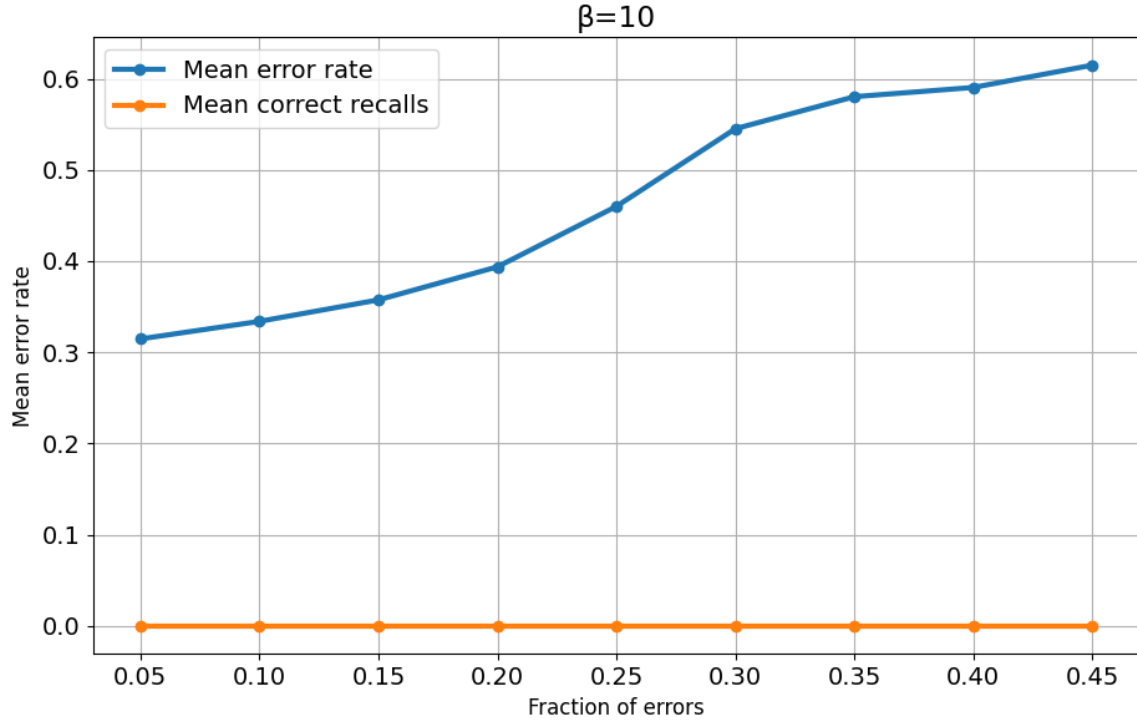


Figure 6: Recall rate and error rate $\beta = 10$

1.4 Qualitative Section

Throughout the simulations of this section we've seen how large β 's improve the retention capacity of the network while large P 's impair it. The theory behind this observation is discussed in the “signal-to-noise” chapter in the book. There, it is shown a network has a limited capacity to retain memories. Moreover we've seen how increasing the value of β allows the network more fixed points which in turn enable it to store more memories. Associative memory is the ability to learn a memory and recall it from a “similar enough” though not necessarily exact same pattern. We can see an example of this in section 1.3, figure 4 where even in cases of 15% dissimilarity in the recall input, the network still manages to output the correct pattern. In this model, learning a new memory will change all other memories as the connectivity matrix would be changed. This does not seem to be the case with “real-life” associative memory. Of course, I might be mistaken as learning this model has changed my perception of past events in which, in fact, I did recall changes in memory as an after effect of learning something new.

2 Modified exam question (2018 Moed A)

2.1 Two-layer feedforward network with N neurons to one output

$$p_i^\mu = \begin{cases} 1 & \text{w.p. } f \\ 0 & \text{w.p. } (1-f) \end{cases}$$

$$q^\mu = \begin{cases} 1 & \text{w.p. } f \\ 0 & \text{w.p. } (1-f) \end{cases}$$

$$w_j = \frac{1}{f(1-f)N} \sum_{\mu=1}^P (q^\mu - f)(p_j^\mu - f), \quad s = H(h - \theta), \quad h = \sum_{j=1}^N w_j r_j$$

We have P patterns ($\mu = 1, \dots, P$), and for some i we have $r_i = p_i^1$.

2.1.1 a. θ For which $P(s = 1 | q^1 = 0) = P(s = 0 | q^1 = 1)$

$$\begin{aligned} h &= \sum_{j=1}^N w_j r_j = \sum_{j=1}^N r_j \frac{1}{f(1-f)N} \sum_{\mu=1}^P (q^\mu - f)(p_j^\mu - f) \\ &= \frac{1}{f(1-f)N} \sum_{j=1}^N \sum_{\mu=1}^P (q^\mu - f)(p_j^\mu - f) r_j \\ &= \frac{1}{f(1-f)N} \left[\sum_{j=1}^N \sum_{\mu=1}^P (q^\mu - f)(p_j^\mu - f) p_j^1 \right] \\ &= \frac{1}{f(1-f)N} \left[\sum_{\mu=1}^P \sum_{j=1}^N (q^\mu - f)(p_j^\mu - f) p_j^1 \right] \\ &= \frac{1}{f(1-f)N} \left[\sum_{j=1}^N (q^1 - f)(p_j^1 - f) p_j^1 + \sum_{\mu=2}^P \sum_{j=1}^N (q^\mu - f)(p_j^\mu - f) p_j^1 \right] \end{aligned}$$

Now we want the expected value of h:

$$\begin{aligned}
E[h|p^1, q^1] &= E \left[\frac{1}{f(1-f)N} \left[\sum_{j=1}^N (q^1 - f)(p_j^1 - f)p_j^1 + \sum_{\mu=2}^P \sum_{j=1}^N (q^\mu - f)(p_j^\mu - f)p_j^1 \right] \right] \\
&= \frac{1}{f(1-f)N} E \left[\sum_{j=1}^N (q^1 - f)(p_j^1 - f)p_j^1 \right] + \frac{1}{f(1-f)N} E \left[\sum_{\mu=2}^P \sum_{j=1}^N (q^\mu - f)(p_j^\mu - f)p_j^1 \right] \\
&= \frac{1}{f(1-f)N} \sum_{j=1}^N (q^1 - f) E \left[(p_j^1 - f)p_j^1 \right] + \frac{1}{f(1-f)N} E \left[\sum_{\mu=2}^P \sum_{j=1}^N (q^\mu - f)(p_j^\mu - f)p_j^1 \right] \\
&= \frac{(q^1 - f)}{f(1-f)N} \sum_{j=1}^N E \left[p_j^1 p_j^1 - f p_j^1 \right] + \frac{1}{f(1-f)N} E \left[\sum_{\mu=2}^P \sum_{j=1}^N (q^\mu - f)(p_j^\mu - f)p_j^1 \right] \\
&= \frac{(q^1 - f)}{f(1-f)N} \sum_{j=1}^N E \left[(1-f)p_j^1 \right] + \frac{1}{f(1-f)N} E \left[\sum_{\mu=2}^P \sum_{j=1}^N (q^\mu - f)(p_j^\mu - f)p_j^1 \right] \\
&= \frac{(q^1 - f)}{f(1-f)N} (1-f)fN + \frac{1}{f(1-f)N} E \left[\sum_{\mu=2}^P \sum_{j=1}^N (q^\mu - f)(p_j^\mu - f)p_j^1 \right] \\
&= (q^1 - f) + \frac{1}{f(1-f)N} E \left[\sum_{\mu=2}^P \sum_{j=1}^N (q^\mu - f)(p_j^\mu - f)p_j^1 \right] \\
&= (q^1 - f) + \frac{1}{f(1-f)N} \sum_{\mu=2}^P \sum_{j=1}^N p_j^1 E \left[(q^\mu - f)(p_j^\mu - f) \right] \\
&= (q^1 - f) + \frac{1}{f(1-f)N} \sum_{\mu=2}^P \sum_{j=1}^N p_j^1 E \left[(q^\mu - f)(p_j^\mu - f) \right] \\
&= (q^1 - f) + \frac{1}{f(1-f)N} \sum_{\mu=2}^P \sum_{j=1}^N E \left[p_j^1 \right] E \left[q^\mu - f \right] E \left[p_j^\mu - f \right] \\
&= (q^1 - f) + \frac{1}{f(1-f)N} \sum_{\mu=2}^P \sum_{j=1}^N E \left[p_j^1 \right] \cdot 0 \cdot 0 \\
&= q^1 - f
\end{aligned}$$

In conclusion - $E[h|p^1, q^1 = 1] = 1 - f$ **and** $E[h|p^1, q^1 = 0] = -f$.

Now we look at the variance:

$$\begin{aligned}
\text{Var}[h|p^1, q^1] &= \text{Var} \left[\frac{1}{f(1-f)N} \sum_{j=1}^N (q^1 - f)(p_j^1 - f)p_j^1 + \frac{1}{f(1-f)N} \sum_{\mu=2}^P \sum_{j=1}^N (q^\mu - f)(p_j^\mu - f)p_j^1 \right] \\
&= \text{Var} \left[\frac{1}{f(1-f)N} \sum_{j=1}^N (q^1 - f)(p_j^1 - f)p_j^1 \right] + \text{Var} \left[\frac{1}{f(1-f)N} \sum_{\mu=2}^P \sum_{j=1}^N (q^\mu - f)(p_j^\mu - f)p_j^1 \right] \\
&= \text{Var} \left[\frac{(q^1 - f)}{f(1-f)N} \sum_{j=1}^N (p_j^1 - f)p_j^1 \right] + \text{Var} \left[\frac{1}{f(1-f)N} \sum_{\mu=2}^P \sum_{j=1}^N (q^\mu - f)(p_j^\mu - f)p_j^1 \right] \\
&= \left(\frac{(q^1 - f)}{f(1-f)N} \right)^2 \sum_{j=1}^N \text{Var} \left[(1-f)p_j^1 \right] + \text{Var} \left[\frac{1}{f(1-f)N} \sum_{\mu=2}^P \sum_{j=1}^N (q^\mu - f)(p_j^\mu - f)p_j^1 \right] \\
&= \left(\frac{(q^1 - f)}{f(1-f)N} \right)^2 N(1-f)^2 \text{Var} \left[p_j^1 \right] + \text{Var} \left[\frac{1}{f(1-f)N} \sum_{\mu=2}^P \sum_{j=1}^N (q^\mu - f)(p_j^\mu - f)p_j^1 \right] \\
&= \left(\frac{(q^1 - f)}{f(1-f)N} \right)^2 N(1-f)^2 \cdot 0 + \text{Var} \left[\frac{1}{f(1-f)N} \sum_{\mu=2}^P \sum_{j=1}^N (q^\mu - f)(p_j^\mu - f)p_j^1 \right] \\
&= 0 + \text{Var} \left[\frac{1}{f(1-f)N} \sum_{\mu=2}^P \sum_{j=1}^N (q^\mu - f)(p_j^\mu - f)p_j^1 \right] \\
&= \left(\frac{1}{f(1-f)N} \right)^2 (P-1) \sum_{j=1}^N (p_j^1)^2 \text{Var} \left[(p_j^\mu - f) \right] \text{Var} \left[(p_j^\mu - f) \right] \\
&= \left(\frac{1}{N} \right)^2 (P-1) \sum_{j=1}^N (p_j^1)^2 \\
&= \left(\frac{1}{N} \right)^2 (P-1) \sum_{j=1}^N p_j^1 \\
&= \frac{(P-1)f}{N} \approx \frac{Pf}{N}
\end{aligned}$$

I.e. $\text{Var}[h|p^1, q^1 = 0] = \text{Var}[h|p^1, q^1 = 1] = \frac{Pf}{N}$. Now we want to choose θ such that it's the average of $E[h|p^1, q^1 = 0]$ and $E[h|p^1, q^1 = 1]$ and as they have the same variance, on average, the amount of times falling to the right of it in one condition is the same as falling left to it in the other so:

$$\theta = \frac{1-f-f}{2} = \frac{1-2f}{2} = \frac{1}{2} - f$$

2.1.2 b. Probability of an error

$$\Pr^- = P(s \neq q^1) \underset{1}{=} P(s = 1|q^1 = 0)f + P(s = 0|q^1 = 1)(1 - f) \underset{2}{=} P(s = 1|q^1 = 0) \underset{3}{=} P(s = 0|q^1 = 1)$$

Where equality #1 is total probability and #2 and #3 are from (a). Now we ask what is $P(s = 0|q^1 = 1)$ or, given that $q^1 = 1$, what is:

$$\Pr \left[\theta < h | q^1 = 1 \right] = ?$$

From article a we know his normally distributed:

$$h \sim N \left(q^1 - f, \frac{Pf}{N} \right) = N \left(\mu, \sigma^2 \right)$$

And so:

$$\Pr \left[\theta < h | q^1 = 1 \right] = \int_{-\infty}^{\frac{\theta - \mu}{\sigma}} \frac{\exp(-z^2)}{\sqrt{2\pi}} dz$$

As

$$\frac{\theta - \mu}{\sigma} = \frac{\frac{1}{2} - f - 1 + f}{\sqrt{\frac{Pf}{N}}} = -\frac{1}{2} \sqrt{\frac{N}{Pf}} = -\sqrt{\frac{N}{4Pf}}$$

We get:

$$\Pr^- = P(s \neq q^1) = \int_{-\infty}^{-\sqrt{\frac{N}{4Pf}}} \frac{\exp\left(\frac{-z^2}{2}\right)}{\sqrt{2\pi}} dz$$

2.1.3 Plug in N, P and f

for $N = 400$, $P = 1000$ and $f = 0.1$ we get

$$\frac{\theta - \mu}{\sigma} = -\sqrt{\frac{N}{4Pf}} = -\sqrt{\frac{400}{4 \cdot 1000 \cdot 0.1}} = -1$$

And according to the table we have:

$$\Pr^- = P(s \neq q^1) = \int_{-\infty}^{-1} \frac{\exp(-z^2)}{\sqrt{2\pi}} dz = 0.16$$

2.2 M copies of the feedforward network

$$s^{(k)} = H \left(\sum_{j=1}^N w_j^{(k)} r_j^{(k)} - \theta \right)$$

$$w_j^{(k)} = \frac{1}{f(1-f)N} \sum_{\mu=1}^P (q^{\mu(k)} - f)(p_j^{\mu(k)} - f)$$

$$s^{\text{out}} = H \left(\frac{1}{M} \sum_{j=1}^M s^{(k)} - \theta^{\text{out}} \right) = H(h^{\text{out}} - \theta^{\text{out}})$$

2.2.1 Find Pr_{out}^-

The probability for $s^{(k)} = 0$ is the probability for $h^{(k)} < \theta$ and so:

$$s^{(k)} | (q^1 = 0) = \begin{cases} 0 & \text{Pr}^- \\ 1 & 1 - \text{Pr}^- \end{cases}$$

and:

$$s^{(k)} | (q^1 = 1) = \begin{cases} 1 & \text{Pr}^- \\ 0 & 1 - \text{Pr}^- \end{cases}$$

Now:

$$\mathbb{E}[h^{\text{out}}] = \mathbb{E} \left[\frac{1}{M} \sum_{j=1}^M s^{(k)} \right] = \frac{1}{M} \sum_{j=1}^M \mathbb{E} [s^{(k)}] = \frac{1}{M} \cdot M \cdot \mathbb{E} [s^{(k)}] = \mathbb{E} [s^{(k)}]$$

So:

$$\mathbb{E} [s^{(k)} | q^1 = 0] = \text{Pr}^- \quad \& \quad \mathbb{E} [s^{(k)} | q^1 = 1] = 1 - \text{Pr}^-$$

Also, note that again, the variance is symmetrical such that:

$$\text{Var}[s^{(k)} | (q^1 = 1)] = \text{Var}[s^{(k)} | (q^1 = 0)] = \text{Pr}^- (1 - \text{Pr}^-)$$

Using the fact that given $q^1 = 0$ or 1, we have $\text{Var} [s^{(k)}] = 0$ and $\text{Cov} [s^{(k)}, s^{(j)}] = 0$ for $1 \neq k \neq j$ (as they are uncorrelated):

$$\text{Var} \left[\frac{1}{M} \sum_{j=1}^M s^{(k)} | (q^1 = 1) \right] = \frac{1}{M^2} \sum_{j=1}^M \text{Var} [s^{(k)} | (q^1 = 1)] = \frac{\text{Pr}^- (1 - \text{Pr}^-)}{M}$$

As we've only used the fact that q^1 is given but did not use its value, we get:

$$\text{Var} \left[\frac{1}{M} \sum_{j=1}^M s^{(k)} | (q^1 = 1) \right] = \text{Var} \left[\frac{1}{M} \sum_{j=1}^M s^{(k)} | (q^1 = 0) \right] = \frac{\text{Pr}^- (1 - \text{Pr}^-)}{M}$$

Just like in the first section (as this is in fact the same problem with different constants), we want to find θ between the two expected values. Again, as they have the same variance, we'd expect the distribution to the left and right of θ would be equal. So:

$$\theta = \frac{\mathbb{E} [s^{(k)} | q^1 = 0] + \mathbb{E} [s^{(k)} | q^1 = 1]}{2} = \frac{\text{Pr}^- + (1 - \text{Pr}^-)}{2} = \frac{1}{2}$$

2.2.2 Calculate Pr_{out}^-

$$P = 1000, N = 400, f = 0.1, M = 25$$

So, given $q^1 = 1$:

$$\frac{\theta - \mu}{\sigma} = \frac{\theta - (1 - \text{Pr}^-)}{\sqrt{\frac{1}{M} \text{Pr}^- (1 - \text{Pr}^-)}} = \frac{0.5 - 1 - 0.16}{\sqrt{\frac{1}{25} \cdot (0.16 - 0.16^2)}} \approx -4.63$$

We then get:

$$\text{Pr}_{\text{out}}^- = \int_{-\infty}^{-4.63} \frac{\exp(-z^2)}{\sqrt{2\pi}} dz \approx 1.6145e-05 = \frac{2.9e-7 + 3.2e-5}{2}$$

2.3 Qualitative Section

The probability of an error due to reading-out from multiple networks rather than just one decreases as the noise averages out over all networks. The networks overlap on the same shared pattern and the rest is uncorrelated noise which is filtered out like the layers were a sieve.